Requirements

Cloud IAM Entitlement Audit

1. Runtime requirements

• Python 3.10 or newer
• Windows, macOS, or Linux
• Local read access to the Terraform or configuration repository being audited

2. Python dependencies

No third-party Python packages are required. The audit script uses only the Python standard library. The accompanying **requirements.txt** is intentionally minimal.

3. Optional security tools

Checkov — useful for broader infrastructure-as-code policy checks. Install with `python -m pip install checkov`, then run `checkov -d . --framework terraform`.

Terraform CLI — optional for formatting and validation. Suggested commands: `terraform fmt -check -recursive` and `terraform validate` from an initialized working directory.

4. Supported inputs

The Python scanner inspects files with these extensions: **.tf, .tfvars, .hcl, .json**. Common generated or vendor directories, including `.git` and `.terraform`, are skipped.

5. Important limitations

This is a local, static, pattern-based scan. It does not connect to AWS, Azure, or Google Cloud; does not evaluate every Terraform expression; and cannot prove that a resource is exploitable or secure. Confirm each result in context and use Checkov and provider-native controls as appropriate.